

Measuring Aerodynamic Properties From Simulation and Experimentation

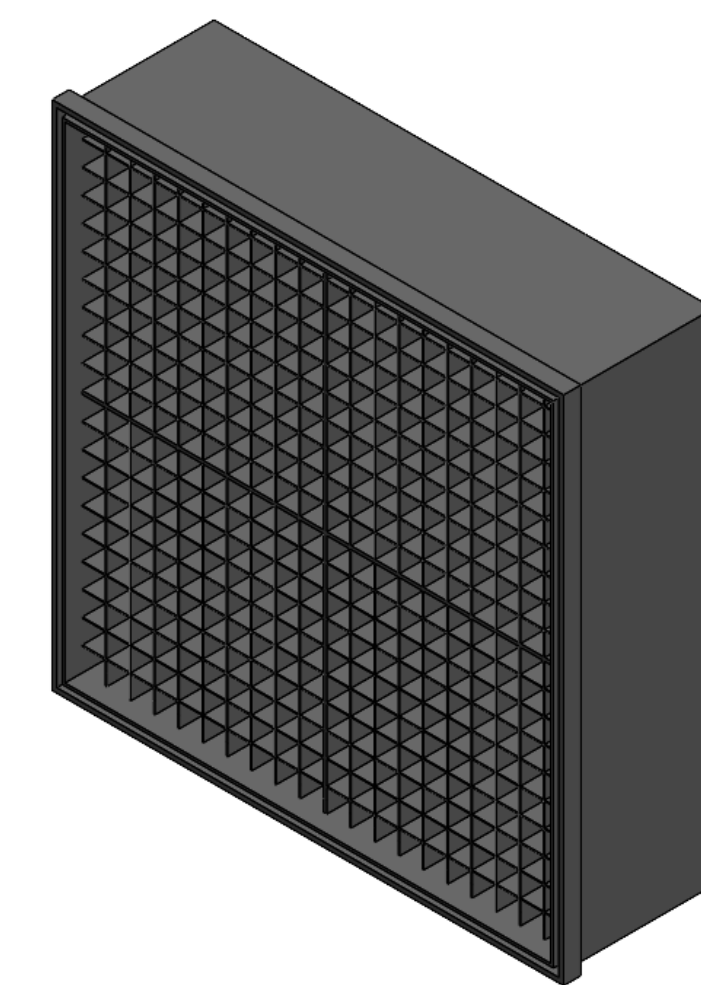
Brycen Schoenberg

Wind Tunnel

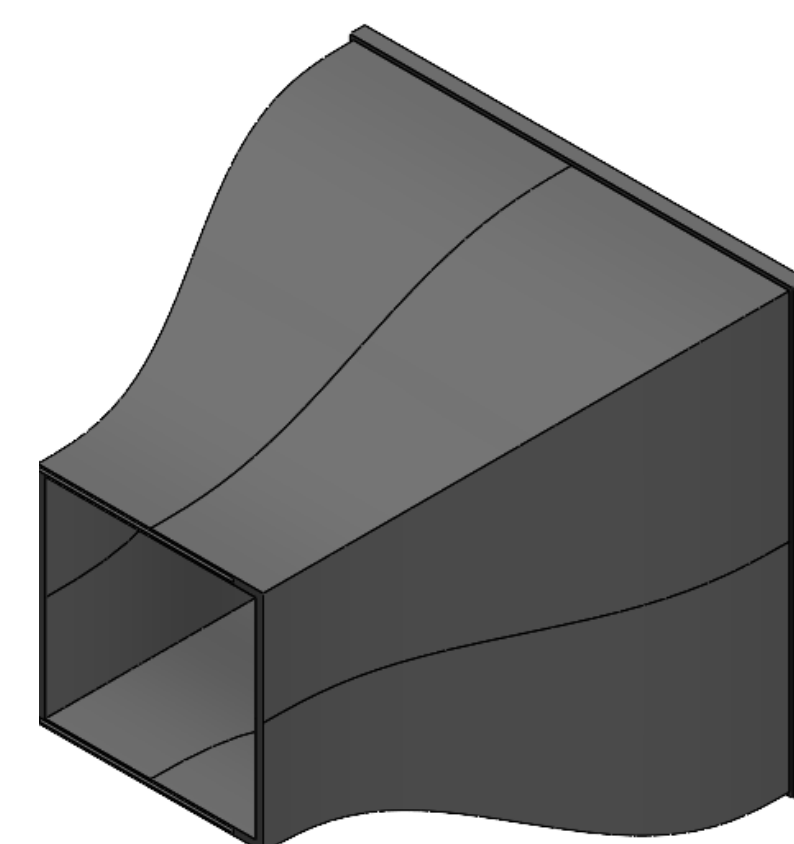
General Idea: Create and design a functional wind tunnel from scratch to measure lift and use it to analyze more complex aerodynamic data.

Components In Order

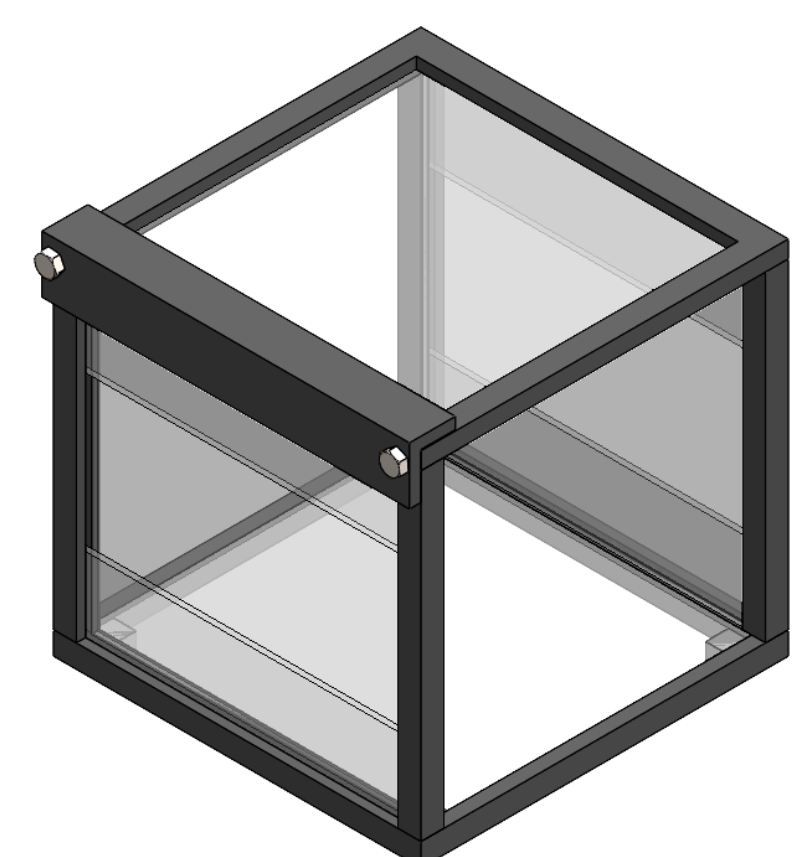
Honeycomb Straightener: Acts as a laminar flow conditioner. It breaks up large-scale atmospheric turbulence and aligns the incoming air into a uniform, parallel stream before it enters the contraction.



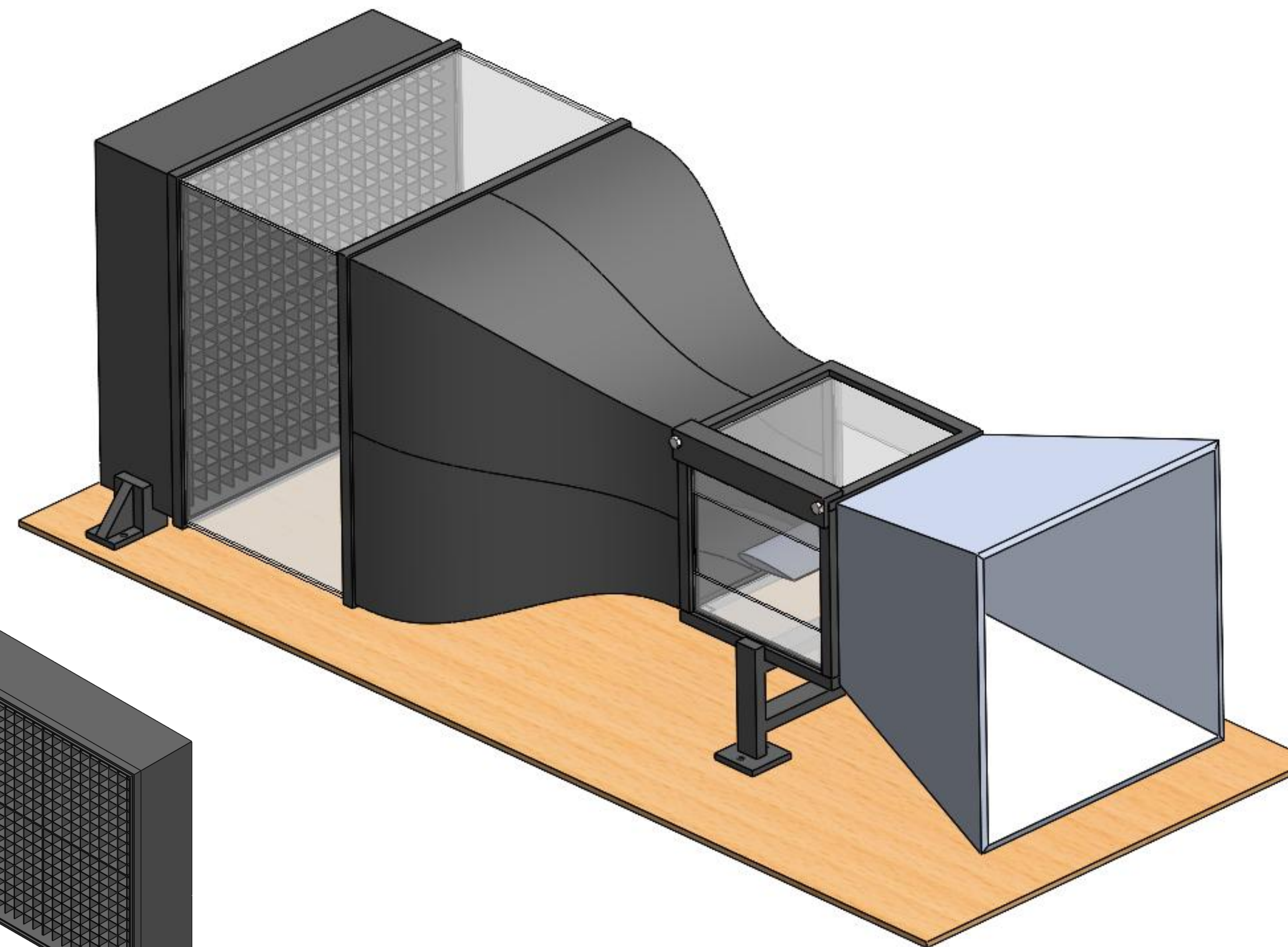
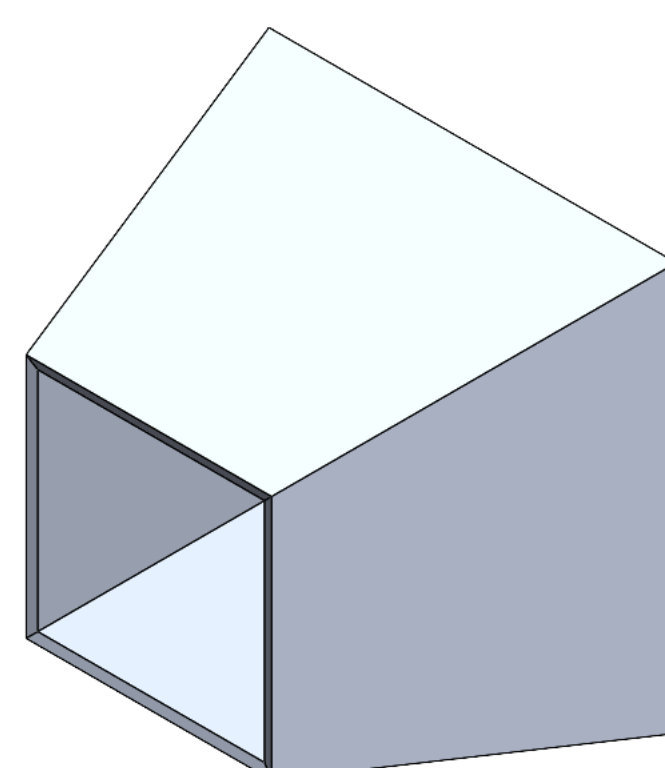
Contraction Chamber: Takes this straightened air and accelerates it through a specific geometry. This increases velocity to more usable values and further reduces turbulence.



Test Chamber: Where the magic happens. Contains the test subject and calibrated sensors to measure aerodynamic properties such as lift, lift coefficient, drag, etc.



Diffuser: Increases cross sectional area to decelerate flow, reduce static pressure and prevent turbulence up stream.



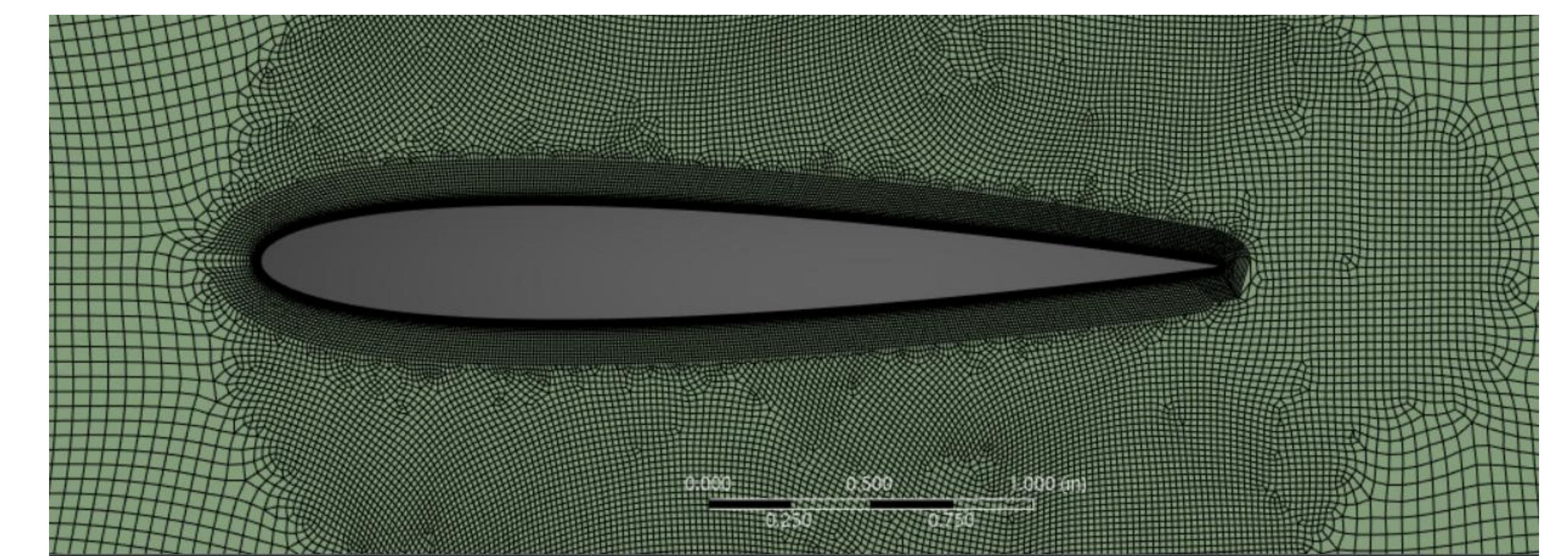
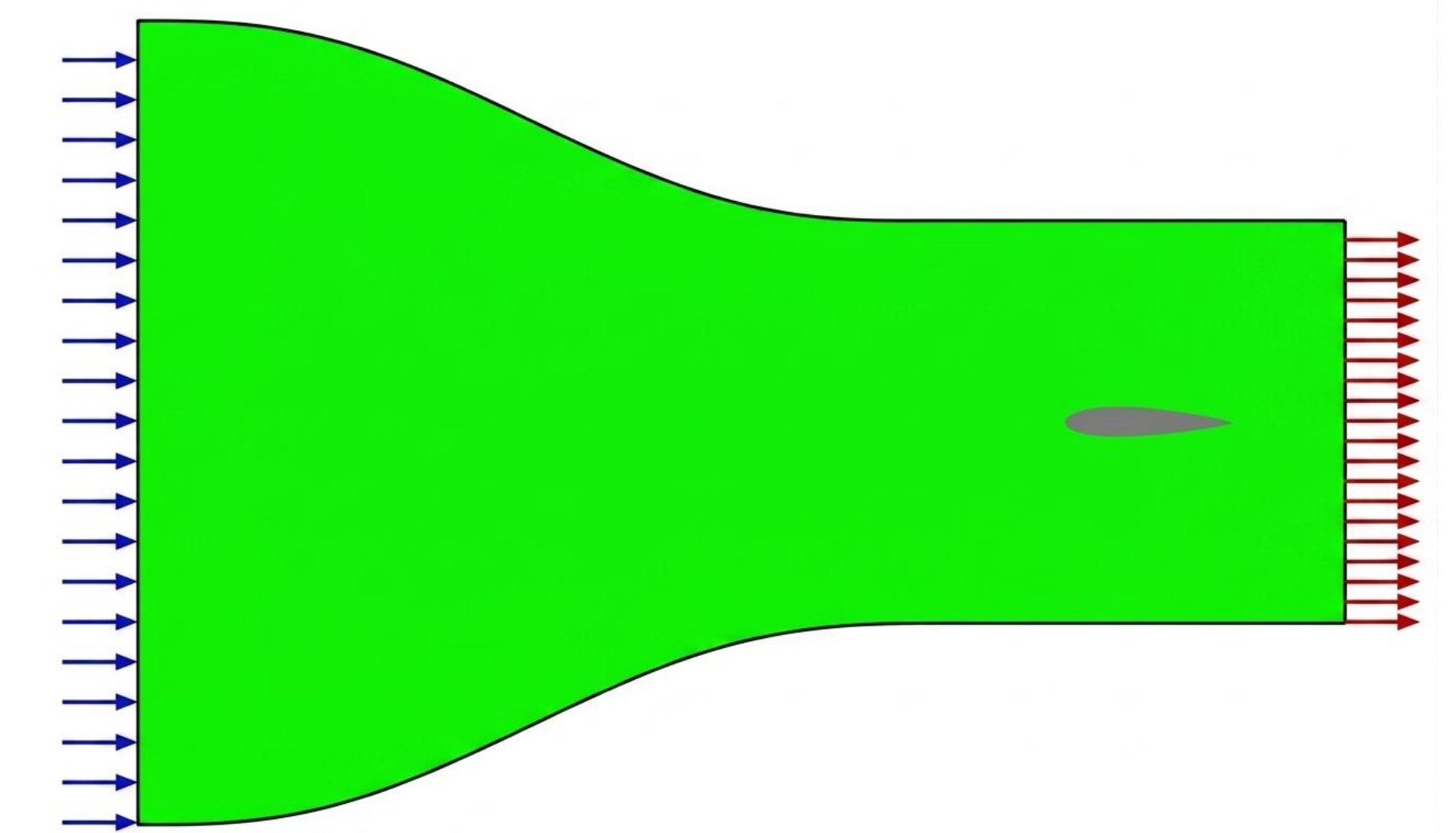
Results for NACA 0012 3" Chord
Airfoil: Coefficient of Lift = 0.0051
Lift = 0.3804 Newtons
Coefficient of Drag = 0.0048

From 2D airfoil theory,
 $Lift = \frac{1}{2}\rho v^2 C_l A$, but $C_l = 0$ at 0 degrees. Lift should be 0. A C_l of 0.0051 is negligible and is considered an accurate representation of data but highlights the impact of a test chamber.

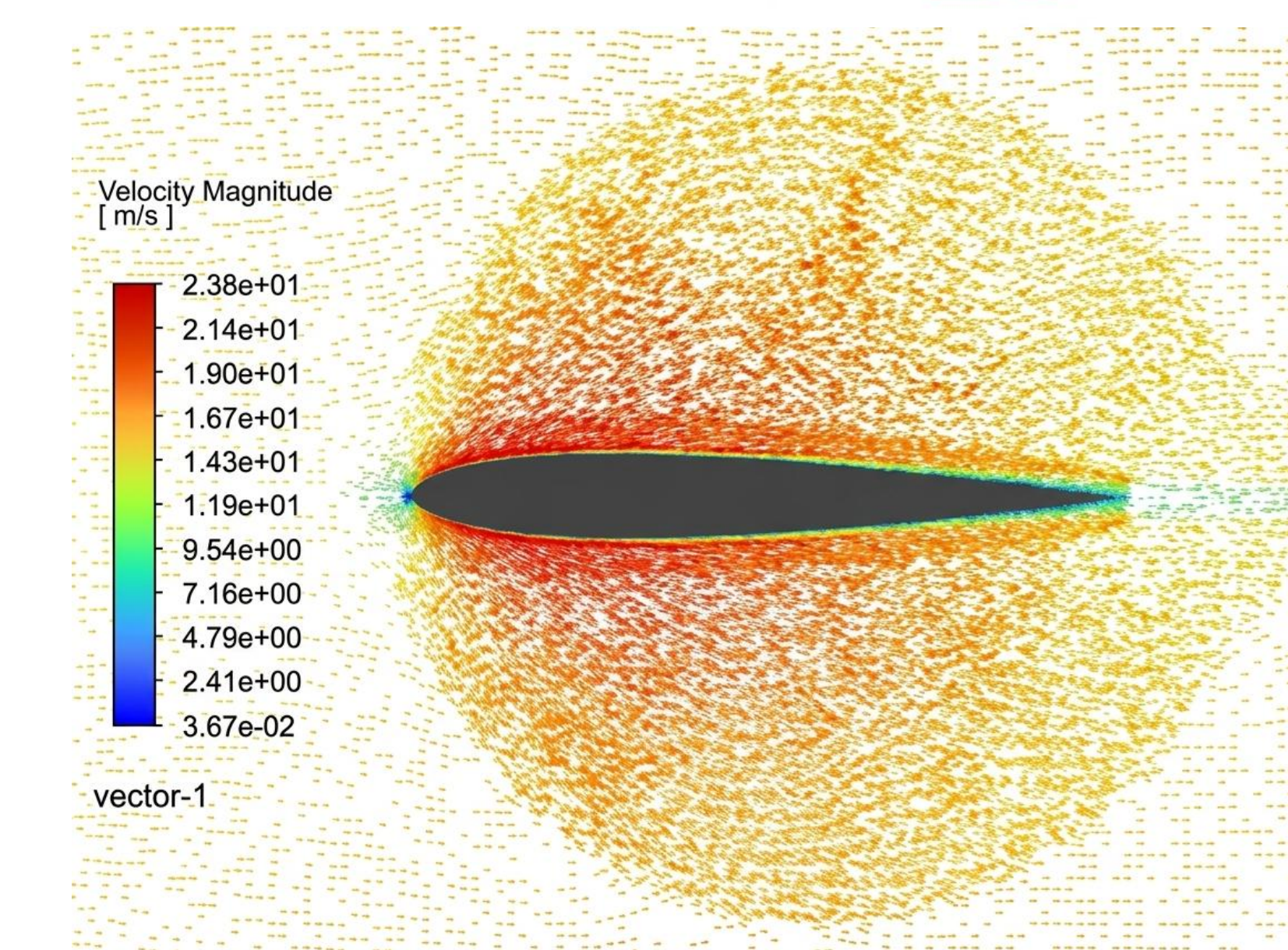
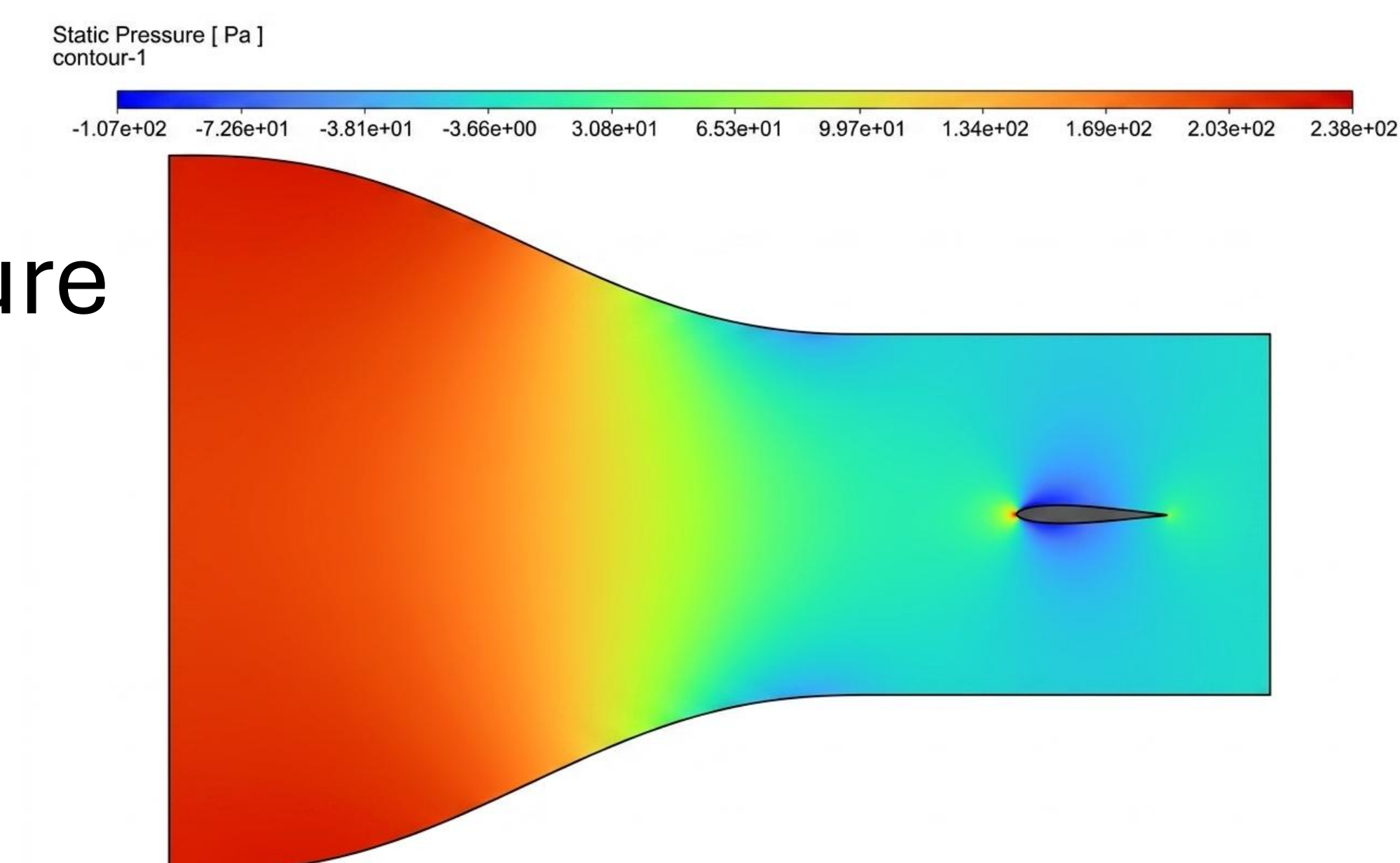
These results will later be compared to wind tunnel results to find data quality and analyze multiple airfoils at different angles.

Simulation

General Idea: Create a 2D replica of the real design and apply virtual conditions that simulate actual wind tunnel behavior. Shown below are the 2D model as well as the mesh used to calculate results.



Plot of pressure distribution across entire tunnel



Plot of velocity vectors over the airfoil